

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes three of the following points: • {complex polysaccharides /they} cannot be digested by human (enzymes) (1) • microorganisms release enzymes that {digest / ferment} (the complex polysaccharides) (1) • release {molecules / glucose/ monosaccharide} (that can be absorbed into the blood and) used in {respiration / metabolism} (1)	accept cellulose/hemicellulose/ pectin/ ignore starch / glycogen accept hydrolysis of glycosidic bond accept named enzyme that digests CHOs eg amylase, cellulase accept {molecules / glucose} are source of energy do not accept produce energy mp3 either for microbes or human	Expert (3)

Question number	Answer	Additional guidance	Mark
8(b)	A description that includes four of the following points: • role of {chemo / baro / thermo} receptors (in the blood) (1) • impulses (from receptors) to {cardiovascular control centre / medulla (oblongata)} (1) • (medulla / brain) sends impulses (to heart / SAN) {via PNS / vagus} (1) • (slowing / reducing) the rate of depolarisation of the SAN (1)	do not accept signal / message unless talking about receptors do not get mps 1 and 2 chemo receptors – increase in CO₂ / decrease in pH baro receptors – blood pressure thermoreceptors – core temperature accept CVC accept reducing rate of contraction	Expert (4)

Question number	Answer	Additional guidance	Mark
8(c)	An answer that includes three of the following points: • (microorganism/T helper cells) release {chemicals/ metabolites} (1) • activating T killer cells (1) • stimulating {clonal expansion/ proliferation} (of T cells) (1) • T killer cells release (chemicals / perforins) which destroy the (tumour) cells (1)	accept cytokines accept cytotoxic T cells accept stimulates mitosis (of T cells)	Expert (3)

Question number	Answer	Additional guidance	Mark
8(d)	A description that includes the following points: • use of fMRI scan {whilst /after} being fed probiotics (1) • to compare (activity) (in regions of the brain controlling memory and sensation) (1)	answers may be in terms of two groups or the same individual accept (named) food containing probiotics description of how active areas can be shown by fMRI scan e.g. active areas will be {lighter / white} accept description of compare	Expert (2)

Question number	Answer	Additional guidance	Mark
8(e)	An answer that includes three of the following points: • (different diet) changes in the gut {flora /bacteria/microbes} (1) • change in the chemicals released (in the gut) (1) • chemicals carried in the blood and passes to the brain through the blood brain barrier (1) • {stimulating / damaging} neurones to release (different) neurotransmitters (1)	accept {stimulation / transmitted} via vagus nerve to brain accept synthesis and availability of {neurotransmitter / serotonin}	Expert (3)

Question number	Answer	Additional guidance	Mark
8(f)	An explanation that includes the following points: • loss of nodes of Ranvier / loss of myelin (1) • slowing the rate of conduction of nerve impulses (1)	accept loss of insulating layer (around neurones) accept loss of Schwann cells accept nerve impulses travel slower accept reduced saltatory conduction accept description of reduced saltatory conduction accept fewer impulses transmitted	Expert (2)

Question number	Answer	Additional guidance	Mark
8(g)	A description that includes three of the following points: • extract (mRNA/LPS receptor gene) from white blood cells (1) • reference to use of (specific) microarray/ DNA probe (1) • collection of data from the microarray analysis (1) • reference to use of bioinformatics to analyse the data (1)	go down both routes and take highest score. Use of cDNA use of PCR gel electrophoresis use of fluorescent {dyes/tags} /DNA probes/ Western blotting / radiolabelled blotting	Expert (3)

